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Element A: Presentation and Justification of the Problem

The problem is that the TV production studio needs new and improved wireless mic holders. They need this to conduct interviews and to speak in the studio and throughout the school while recording the WHS Timber TV news or Ambush News. This is important because TV production students want to hold an authentic looking microphone, as seen on TV, as opposed to the actual microphone transmitter.¹

Current available solutions, such as previous designs by other students, do not hold the microphone transmitter when moved wildly.² High school students in the TV production class need to use these microphones to interview their fellow students.³ They interview students during news segments, student interviews, and surprise interviews called Ambush News.

Our group will design a comfortable, authentic, and secure microphone holder that holds the microphone transmitter at every orientation and resembles a real microphone with quantifiable constraints such as weighing less than 2 pounds and being less than 3x3x15 inches in volume. Previous renditions of the holder did not work well because they did not hold the transmitter⁴ and they did not look authentic enough. Solving this problem would be beneficial as there are not many other similar products, so if our group were to develop a functioning product we would establish our holder as the premiere microphone holder design. The microphones are used in every episode of the news and are fundamental to anything done in the television studio.⁵



(1)

Figure 1: School news department (ambush news segment) using the transmitter without any microphone holder (2023 HS Student)

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(2)

Figure 2: old mic holder had the transmitter stick too far out

Figure 1: TV, WHS Timber. "Ambush News 01 2023 Griff Grady." *YouTube*, YouTube, 31 Jan. 2023,
www.youtube.com/watch?v=jF-1oVaruGM&list=PL_4r8ioYvuM6myQBdu88KkSiPyJBqCFgf&index=3.

Figure 2: (2023 HS Student)

Stakeholders-

- 1: Peter O'Farrell - Walpole High School TV Production teacher; main stakeholder who asked for project
- 2: TV Production Students - students of Mr O'Farrell's TV Production class; need mics to record news
- 3: James Connolly - Walpole High School Film Festival Advisor/Work Study Coordinator
- 4: Walpole High schoolers getting interviewed - want to be interviewed like on TV

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This problem needs to be solved now so that the TV production students can wrap up the year with good mics that work well and look good. They also will need it for events such as graduation and the end of the year and all of the winter and spring sports and concerts.

Works Cited

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